Complex Numbers and Operations

Name _____

Algebra 2 Lesson 2-4 (fast track) Date _____ Period _____

Learning goal: I can multiply complex numbers and write each product in the form $a + bi$.

Language goal: I can explain each step of a product using the words *imaginary unit*, *real part*, *imaginary part*, and *complex conjugates*.

On the quiz: (Savvas numbers) (1) Solve $x^2 = -64$ over the complex numbers (full Savvas quiz only). (2) Bonus: subtract two complex numbers; combine the real parts, then the imaginary parts (Practice b). (3) Multiply a complex number by its *complex conjugate*: the i terms cancel and the product is a real number (Example 3B, Try It 3b). (4) Multiply three factors (an imaginary number times two binomials) and write the answer as $a + bi$ (Example 3A and 3B, Practice a). (5) Match products with their complex factors (full Savvas quiz only).

On the Topic 2 test: Question 9: multiply a number by its conjugate, $(4 + i)(4 - i)$, as $a + bi$ (Example 3). Question 8 (solve $x^2 = -25$) comes from Example 1, which this packet skips.

Complex numbers. The **imaginary unit** i is the principal square root of -1. Then $i^2 = -1$. A **complex number** can be written in the form $a + bi$, where a and b are real numbers: a is the **real part**, b is the **imaginary part**, and bi is the imaginary term. For example: $-6 + 4i$ ($a = -6$, $b = 4$), $7 - i\sqrt{2}$ ($a = 7$, $b = -\sqrt{2}$), $0.5i$ ($a = 0$, $b = 0.5$).

Spanish: imaginary unit = *unidad imaginaria*; complex number = *número complejo*; complex conjugates = *conjugados complejos*.

Example 3: Multiply Complex Numbers

How can you write each product in the form $a + bi$? **The idea:** distribute as you would with binomials, replace every i^2 with -1, then write the real part first.

A. $-2.5i(8 - 9i)$

$$\begin{aligned} -2.5i(8 - 9i) &= (-2.5i)(8) + (-2.5i)(-9i) \\ &= -20i + 22.5i^2 \\ &= -20i + 22.5(-1) \quad (i^2 = -1) \\ &= -22.5 - 20i \end{aligned}$$

The product is $-22.5 - 20i$ (real part first).
Watch out: $22.5i^2$ is -22.5, not 22.5.

B. $(3 - 2i)(3 + 2i)$

$$\begin{aligned} (3 - 2i)(3 + 2i) &= 3(3 + 2i) - 2i(3 + 2i) \\ &= 9 + 6i - 6i - 4i^2 \\ &= 9 + 6i - 6i - 4(-1) \\ &= 13 \end{aligned}$$

The product is 13, a real number: $3 - 2i$ and $3 + 2i$ are **complex conjugates** (same real part, opposite imaginary parts), so $6i - 6i = 0$.

Complex conjugates: $(a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 - b^2(-1) = a^2 + b^2$, always a real number.

Say it: In Part A, I replaced i^2 with _____ because the *imaginary unit* squared is _____. The *real part* of the product is _____ and the *imaginary part* is _____. In Part B, $3 - 2i$ and $3 + 2i$ are *complex conjugates*, so the i terms _____ and the product is _____.

Try It 3: Write each product in the form $a + bi$

a. $\frac{2i}{5}\left(10 - \frac{5i}{2}\right)$

Product: _____

b. $\left(\frac{1}{2} + 2i\right)\left(\frac{1}{2} - 2i\right)$

Product: _____

In part b, why is there no imaginary part in the answer? Use the words *complex conjugates*.

Practice: Get ready for quiz questions 4 and 2

a. Write the expression $2i(3 - 4i)(1 + i)$ in the form $a + bi$.
Multiply the two binomials first (like Part B), then multiply by $2i$ (like Part A).

Answer: _____ + _____ i

b. What is the difference of $(4+2i)$ and $(1+7i)$?
Distribute the negative to *both* parts first:
$(4 + 2i) + (-1 - 7i)$.

Answer: _____

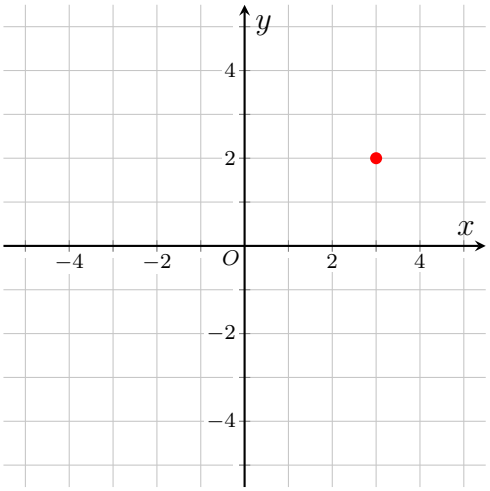
IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
VZ8 ★	Multiply complex numbers	Quiz 3, 4: distribute, $i^2 = -1$ (Ex. 3)
7U5 ★	Complex conjugates	Quiz 3: times its conjugate (Ex. 3B)
JVF ★	Add and subtract complex numbers	Quiz 2, the +5 bonus (Practice b)
5VV	Introduction to complex numbers	Warm-up (optional): i and $a + bi$
FG7	Solve a quadratic equation using square roots	Quiz 1 (full Savvas quiz only); optional

Exit question: Multiply by i (DOK 3)

The complex number $a + bi$ can be represented on a coordinate plane as the point (a, b). You can use multiplication by i to rotate a point about the origin in the coordinate plane.



a. Write the point (x, y) on the graph as the complex number $x + yi$.

_____

b. Multiply the complex number by i. Interpret the new value as a new point in the plane.

_____

_____

c. Repeat the steps above for two other points. How does multiplication by i rotate a point?

_____

_____

_____

Before the quiz, finish these sentences.

To multiply complex numbers, I distribute, then replace i^2 with _____ because i is the *imaginary unit*.

$4 + 5i$ and $4 - 5i$ are *complex conjugates*, so their product has no _____.

In $-22.5 - 20i$, the *real part* is _____ and the *imaginary part* is _____.